validating models. However, one limitation is the lack of standardization in data collection, which reduces their usability. Each dataset di ers in structure, semantics, and

EV-insights framework design

EV-Insights is a framework aimed at optimizing the management of EV charging data. is section o ers a detailed overview of the system architecture, along with a descrip-

- Admin: handles database initialization, and administrative tasks such as cleaning or resetting the database.

Services

is section presents a comprehensive overview of the various functionalities integrated

Figure 4 illustrates an example based on the ACN Caltech dataset. e data reveals

tail distribution. is insight shows typical charging behaviors for electric vehicles in the Norway_12loc dataset.

plug durations from the Harvard_Dataverse dataset. Most sessions are shorter than 4 hours, with a clear peak during the rst two hours. e distribution declines rapidly for longer durations, indicating that short or partial charging sessions are more common.

- Plug duration by plug-in hour: examines the relationship between plug durations and the hour of the day to illustrate how plug durations vary across di erent hours.

- XGBoost Charge Duration: predicts the charge duration of the next charging session of a speci c user. Predictions are provided by using the XGBoost [

Ingesting data into a database

Single user

DSO, charging station owners, private users, or any interested party can leverage EV-

illustrating simple statistics computed for an individual user and for an entire dataset, respectively.

Additionally, EV-Insights can provide insights into the user's charging behavior over